Chapter 1

Cross-Spectra analysis and Cosmological parameters

Contents

Introduction

The previous chapters focused on map-domain approaches to the analysis of QUBIC observations. In particular, the Frequency Map-Making (FMM) algorithm provides a set of reconstructed frequency maps, while the Component Map-Making (CMM) framework performs component separation directly during the map reconstruction process. However, we have not yet described how component separation can be performed using frequency maps alone. This is achieved through the widely used *cross-spectrum analysis* method. This approach relies on the spectral behaviour of the observed sky. Since the Cosmic Microwave Background (CMB), thermal dust, synchrotron emission, and other astrophysical foregrounds exhibit distinct frequency dependencies, their contributions can be statistically disentangled by exploiting the correlations between maps observed at different frequencies. Rather than reconstructing individual component maps, the analysis is performed directly on the angular power spectra and cross-spectra derived from the reconstructed frequency maps. Cross-spectrum methods offer several advantages. First, they naturally suppress noise biases when independent noise realisations are available, typically by splitting the data into noise independent subsets (for example different detectors, forward and backward scans, or different observing days)[1]. Second, they provide a flexible framework for modelling astrophysical foregrounds and instrumental effects directly at the statistical level. Finally, they establish a direct

[1]The spectral imaging capabilities of QUBIC may complicate such data splits, as they introduce noise correlations between reconstructed sub-bands. This effect is still under investigation.

connection between the measured observables and the theoretical predictions of cosmological models.

The objective of this chapter is therefore twofold. The first part presents the construction and modelling of the multi-frequency angular power spectra obtained from FMM frequency maps, together with the methodology used to separate the different astrophysical contributions. The second part describes how these spectra can be used to constrain cosmological parameters, with particular emphasis on the tensor-to-scalar ratio r, which constitutes the primary scientific target of QUBIC.

Personal contribution: The cross-spectrum component separation framework and the associated cosmological parameter estimation pipeline were originally developed jointly with Mathias Regnier for the methodological studies presented in [1, 2], as well as for his PhD thesis. During the second and third years of my PhD, I significantly extended these tools by improving their generality, software architecture, and integration within the QUBIC simulation and reconstruction framework. I also automated large parts of the analysis pipeline and introduced plotting routines, enabling the production of end-to-end forecasts with minimal manual intervention.

1.1 Cross-Spectrum component separation

1.1.1 General Principle

The general idea of cross-spectrum analysis is to compare angular cross-power spectra between frequency maps in order to constrain both foreground and CMB parameters. Given a physical emission model for each astrophysical component, one can derive its associated angular power spectrum and its frequency dependence, and thus predict the corresponding cross-spectra. The availability of multiple frequency channels improves the constraints on these parameters, making this approach particularly well suited to QUBIC, which can reconstruct multiple sub-bands within its bandwidth through spectral imaging. We call cross-spectrum the angular cross-power spectrum between two frequency maps, noted $D_\ell^{\nu_i \times \nu_j}$, and auto-spectrum the angular power spectrum of one frequency map, noted $D_\ell^{\nu_i \times \nu_i} \equiv D_\ell^{\nu_i}$.

Cross-spectra model

For a sky composed of CMB, thermal dust, and synchrotron emission, the cross-spectrum between two frequency maps at ν_i and ν_j can be written as the sum of the individual contributions:

$$
\begin{aligned}
\mathcal{D}_\ell^{\nu_i\times\nu_j} &= \mathcal{D}_{\ell,CMB}^{\nu_i\times\nu_j} + \mathcal{D}_{\ell,Dust}^{\nu_i\times\nu_j} + \mathcal{D}_{\ell,Sync}^{\nu_i\times\nu_j} + \mathcal{D}_{\ell,Dust\times Sync}^{\nu_i\times\nu_j} \\
&= r\times\mathcal{D}_\ell^{\text{tensor}} + A_{\text{lens}}\times\mathcal{D}_\ell^{\text{lensed}} \\
&+ A_d\Delta_d f_d^{\nu_i}(\beta_d,T_d) f_d^{\nu_j}(\beta_d,T_d)\left(\frac{\ell}{\ell_0}\right)^{\alpha_d} + A_s\Delta_s f_s^{\nu_i}(\beta_s,T_s) f_s^{\nu_j}(\beta_s,T_s)\left(\frac{\ell}{\ell_0}\right)^{\alpha_s} \\
&+ \varepsilon\sqrt{A_dA_s}\left(f_d^{\nu_i}(\beta_d,T_d)f_s^{\nu_j}(\beta_s,T_s) + f_s^{\nu_i}(\beta_s,T_s)f_d^{\nu_j}(\beta_d,T_d)\right)\left(\frac{\ell}{\ell_0}\right)^{\frac{\alpha_d+\alpha_s}{2}}.
\end{aligned}
\tag{1.1}
$$

This expression combines the contributions of the CMB (tensor and lensed scalar modes), thermal dust, and synchrotron emission, as well as a possible spatial correlation term between dust and synchrotron components. Each foreground contribution depends on its spectral energy distribution $f^{\nu}_{\text{comp}}(\beta_{\text{comp}})$, as introduced in Section 1.4, and is normalised at a reference multipole ℓ_0, typically set to 80 [3].

The CMB B-modes contribution depends on two parameters, fixing the other parameters to Planck's best fit [4]: the tensor-to-scalar ratio r, which is the primary scientific target of QUBIC, and the gravitational lensing amplitude A_{lens}. In the simulations, r is set to zero, while A_{lens} is fixed to unity, since QUBIC is not expected to have sufficient sensitivity to constrain lensing effects (see subsection 1.4.1). The thermal dust component depends on several parameters. These include the dust amplitude A_d, defined at a reference frequency $\nu_{0,d} = 353$ GHz, corresponding to the most dust-dominated polarised Planck channel; the angular power-law index α_d; the spectral index β_d; the dust temperature T_d; and the frequency decorrelation parameter Δ_d. Following [5], the dust temperature is fixed to $T_d = 20$ K and the spectral index to $\beta_d = 1.54$ in our simulation. In addition, the decorrelation parameter Δ_d is not independently fitted, as it is fully degenerate with the amplitude A_d. The synchrotron component is described by an analogous set of parameters, with a reference frequency fixed at $\nu_{0,s} = 23$ GHz. It is characterised by its amplitude A_s, spectral index β_s, angular power-law index α_s, and frequency decorrelation parameter Δ_s, following the same formalism as for dust. Finally, the spatial correlation between dust and synchrotron emissions is parametrised by ε.

Cross-spectra computation

The cross-spectra are computed from the reconstructed frequency maps using the *NaMaster* package [6][2], which is widely used for pseudo-C_ℓ estimation in CMB analyses. The pseudo-angular power spectra are estimated from HEALPix [7] maps with a resolution parameter $N_{\text{side}} = 256$. The multipole range considered extends from $\ell_{\min} = 40$ to $\ell_{\max} = 2N_{\text{side}} - 1 = 511$, with a constant bin width Δ_ℓ. The upper limit is imposed by the pseudo-C_ℓ estimator implemented in NaMaster, whose accuracy degrades for polarisation maps at multipoles significantly larger than $2N_{\text{side}}$. The lower limit is determined based on the angular size of the observing patch, which is 15 °. This angular size corresponds to 1.7% of the sky, and the associated angular scale is $\ell = 12$.

[2] `https://github.com/LSSTDESC/NaMaster`

But, to have enough modes[3] within each bin, it has been observed empirically that $\ell_{\min} = 40$ provides sufficient sampling. Finally, the binning parameter Δ_ℓ was determined empirically as a compromise between spectral resolution and statistical uncertainty. Larger bins reduce the variance of the estimated spectra, while smaller bins preserve more information on their multipole dependence. Spectrum binning is also performed by NaMaster to deconvolve the mode-coupling introduced by observing a partial sky. Masking the sky breaks the orthogonality of spherical harmonics, coupling power between neighbouring multipoles; binning the spectra yields an invertible mode-coupling matrix, allowing this coupling to be corrected for, and therefore transforming the measured pseudo-C_ℓ into actual C_ℓ.

1.1.2 External data

As the frequency range covered by QUBIC is limited to $[130, 245]$ GHz, it can be advantageous to incorporate external data providing information at frequencies outside this interval. The inclusion of such data increases the spectral leverage available for component separation, improving the discrimination between the different astrophysical contributions. In the case of a sky model containing the CMB and Galactic dust emission, the addition of the Planck 353 GHz channel is particularly valuable, as this frequency is highly sensitive to dust emission.

Beyond extending the frequency coverage, external data also increase the number of frequency bands available for the cross-spectra analysis. Since the likelihood is built from all auto- and cross-power spectra between frequency bands, the total number of spectra scales as :

$$N_{\text{spec}} = \frac{N_{\text{freq}}\,(N_{\text{freq}} + 1)}{2}. \tag{1.2}$$

Adding external frequency channels therefore increases the number of spectra entering the likelihood, providing additional constraints on the model parameters. From this perspective, it can also be beneficial to include frequency bands that overlap with those already observed by QUBIC. For instance, the Planck 143 GHz and 217 GHz channels can be incorporated alongside the QUBIC bands. Although these frequencies are already covered by QUBIC, they contribute additional auto- and cross-spectra, thereby increasing the amount of information available for parameter estimation.

1.1.3 Noise debiasing

The power spectrum of the reconstructed maps can be written as the sum of the signal power spectrum, noted S_ℓ, and the noise power spectrum, N_ℓ. To remove the bias introduced by N_ℓ during the parameters' estimation, we will perform multiple end-to-end simulations while varying the noise seed. Then, we compute the average cross-power spectrum of the reconstructed maps $\overline{D_\ell}^{\nu_i \times \nu_j}$ and the average cross-power spectrum of the residual maps, which is equivalent to $\overline{N_\ell}^{\nu_i \times \nu_j}$. By simply computing the difference between the two, we can compute the unbiased cross-power spectrum of the signal $\overline{S_\ell}^{\nu_i \times \nu_j} \equiv \overline{D_\ell}^{\nu_i \times \nu_j} - \overline{N_\ell}^{\nu_i \times \nu_j}$. The Figure 1.1 shows the unbiasing process. The

[3] The effective number of modes can be computed through : $N_{\text{modes}} = f_{\text{sky}} \sum_{\ell \in \text{bin}} (2\ell + 1)$.

diagonal plots are the auto-spectra, while the off-diagonal term is the cross-spectrum between the two frequency maps. The orange points are $\overline{D_\ell}^{\nu_i\times\nu_j}$, with contribution from sky signals and noise bias, while blue points are the unbiased cross-spectra $\overline{S_\ell}^{\nu_i\times\nu_j}$. We also added the theoretical model for cross-power spectra, based on equation 1.1 for a sky composed of dust and CMB, with the red dashed line. The orange line completely drifts away from the model, while the blue line (zoomed in the right figure) is correctly centred around it. These spectra are computed from two frequency maps using the UWB design, meaning that spectral imaging is used to split the large band into the two sub-bands. This splitting introduces anti-correlation, as discussed in Chapter 4, which are visible looking at the off-diagonal spectrum $\overline{D_\ell}^{150\times220}$.

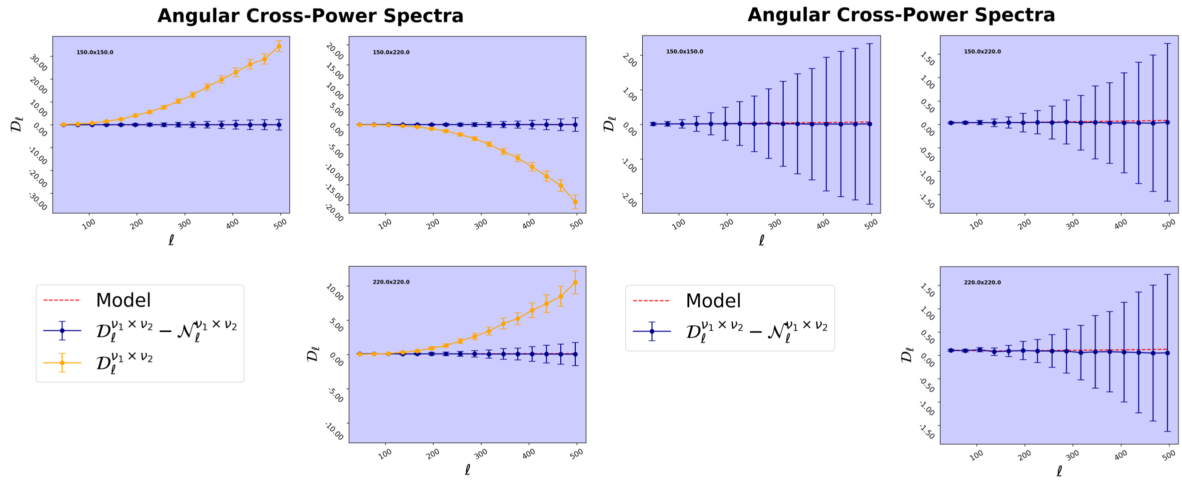

Figure 1.1: Comparison between biased and unbiased BB cross-power spectra computed over two frequency maps. (left) The orange points are spectra containing sky signal and noise, while blue points are the spectra where we removed the noise bias. Red dash line shows the theoretical model for the signal spectra. (right) Same figure without the biased spectra, to show that the results are indeed centred around the model.

1.1.4 Noise covariance matrix

To estimate the foreground and CMB parameters, we construct a noise covariance matrix that is used to weight the cost function entering the fitting procedure. Several contributions are included in this covariance matrix. The first is the covariance of the noise realisations, $\mathrm{Cov}(N_\ell^{\mathrm{sim}})$, which is required to correctly account for the uncertainty on the measured power spectra and to debias the signal spectra. An example of the corresponding correlation matrix is shown in Figure 1.2.

Figure 1.2 displays the correlation matrix associated with the set of auto- and cross-spectra formed from the QUBIC 150 and 220 GHz bands together with the Planck 353 GHz channel. The matrix is organised into blocks of size $(N_{\mathrm{bin}}, N_{\mathrm{bin}})$, where N_{bin} is the number of multipole bins. Diagonal blocks correspond to auto-spectra, while off-diagonal blocks correspond to cross-spectra between different frequency bands. The colour scale represents the correlation coefficient, with red indicating positive correla-

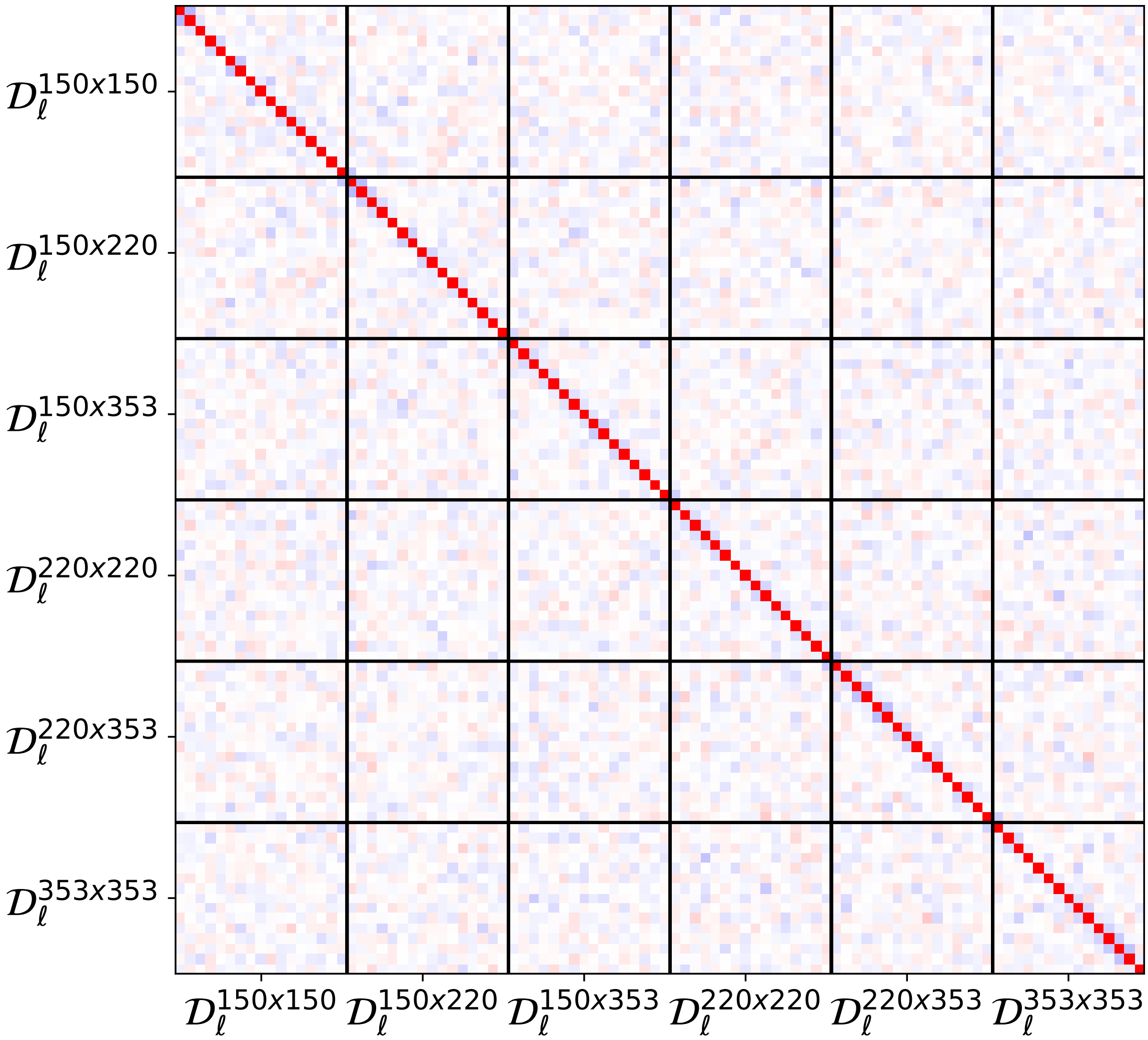

Figure 1.2: Correlation matrix of the cross-spectra obtained from the 150 and 220 GHz bands of QUBIC using the DB design, and the 353 GHz band of Planck. Red corresponds to a correlation coefficient of +1, blue to −1, and white to 0. Figure taken from [1]. It is important to note that with spectral imaging, anti-correlations would be visible between QUBIC's frequency sub-bands.

tions, blue indicating negative correlations, and white indicating negligible correlations. In this example, the DB design (see Section 2.4.1) is used, meaning that no spectral imaging is performed. Consequently, the QUBIC 150 and 220 GHz maps are reconstructed independently and exhibit no correlation with each other, which is reflected in the corresponding off-diagonal blocks of the matrix.

Additionally, the statistical uncertainty associated with the observation, the *sample variance*, must be taken into account. In general statistical terms, sample variance refers to the uncertainty arising when a random quantity is estimated from a finite number of realisations or measurements. In the specific context of CMB analysis, several contributions can be distinguished. The first, intrinsic contribution is the *cosmic variance*, which arises from the fact that the CMB is a Gaussian random field and that we observe only a single realisation of it. For the CMB angular power spectrum, the corresponding expression for cosmic variance is given in Eq. 1.18. Additional contributions to the sample variance arise from the limited sky coverage of the observation, which reduces the number of accessible modes, as well as from the binning of the angular power spectrum in multipole space.

Therefore, we need to compute the covariance between the cross-spectra. First, we consider the cross-spectrum estimator :

$$\mathcal{C}_\ell^{\nu_i \times \nu_j} = \frac{1}{2\ell+1} \sum_{m=-\ell}^{\ell} a_{\ell m}^{\nu_i} \, a_{\ell m}^{\nu_j\,*}. \tag{1.3}$$

The covariance between two cross-spectra is defined as :

$$\mathrm{Cov}\Big(\mathcal{C}_\ell^{\nu_i \times \nu_j}, \mathcal{C}_{\ell'}^{\nu_k \times \nu_l}\Big) = \big\langle \mathcal{C}_\ell^{\nu_i \times \nu_j} \, \mathcal{C}_{\ell'}^{\nu_k \times \nu_l} \big\rangle - \big\langle \mathcal{C}_\ell^{\nu_i \times \nu_j} \big\rangle \big\langle \mathcal{C}_{\ell'}^{\nu_k \times \nu_l} \big\rangle. \tag{1.4}$$

Expanding the estimators gives :

$$\begin{aligned}\mathrm{Cov}\Big(\mathcal{C}_\ell^{\nu_i \times \nu_j}, \mathcal{C}_{\ell'}^{\nu_k \times \nu_l}\Big) = \frac{1}{(2\ell+1)(2\ell'+1)} \sum_{m=-\ell}^{\ell} \sum_{m'=-\ell'}^{\ell'} & \Big(\big\langle a_{\ell m}^{\nu_i} a_{\ell m}^{\nu_j\,*} a_{\ell' m'}^{\nu_k} a_{\ell' m'}^{\nu_l\,*} \big\rangle \\ & - \big\langle a_{\ell m}^{\nu_i} a_{\ell m}^{\nu_j\,*} \big\rangle \, \big\langle a_{\ell' m'}^{\nu_k} a_{\ell' m'}^{\nu_l\,*} \big\rangle \Big).\end{aligned} \tag{1.5}$$

Assuming Gaussianity, Wick's theorem [8] gives :

$$\begin{aligned}\big\langle a_{\ell m}^{\nu_i} a_{\ell m}^{\nu_j\,*} a_{\ell' m'}^{\nu_k} a_{\ell' m'}^{\nu_l\,*} \big\rangle = & \big\langle a_{\ell m}^{\nu_i} a_{\ell m}^{\nu_j\,*} \big\rangle \, \big\langle a_{\ell' m'}^{\nu_k} a_{\ell' m'}^{\nu_l\,*} \big\rangle \\ & + \big\langle a_{\ell m}^{\nu_i} a_{\ell' m'}^{\nu_k} \big\rangle \, \big\langle a_{\ell m}^{\nu_j\,*} a_{\ell' m'}^{\nu_l\,*} \big\rangle \\ & + \big\langle a_{\ell m}^{\nu_i} a_{\ell' m'}^{\nu_l\,*} \big\rangle \, \big\langle a_{\ell m}^{\nu_j\,*} a_{\ell' m'}^{\nu_k} \big\rangle \, .\end{aligned} \tag{1.6}$$

The first term cancels the disconnected part in the covariance definition. For a statistically isotropic field, the harmonic coefficients satisfy :

$$\langle a_{\ell m}^{\nu_a} a_{\ell' m'}^{\nu_b\,*} \rangle = \mathcal{C}_\ell^{\nu_a \times \nu_b} \, \delta_{\ell \ell'} \, \delta_{m m'}. \tag{1.7}$$

Using this, only the cross-pairings survive, yielding :

$$\mathrm{Cov}\Big(\mathcal{C}_\ell^{\nu_i\times\nu_j},\mathcal{C}_{\ell'}^{\nu_k\times\nu_l}\Big) = \frac{1}{(2\ell+1)(2\ell'+1)}\sum_{m,m'}\delta_{\ell\ell'}\,\delta_{mm'}\Big(\mathcal{C}_\ell^{\nu_i\times\nu_k}\,\mathcal{C}_\ell^{\nu_j\times\nu_l}+\mathcal{C}_\ell^{\nu_i\times\nu_l}\,\mathcal{C}_\ell^{\nu_j\times\nu_k}\Big). \tag{1.8}$$

The Kronecker deltas enforce $\ell = \ell'$ and $m = m'$, so that :

$$\sum_{m,m'}\delta_{mm'} = \sum_{m=-\ell}^{\ell} 1 = 2\ell+1. \tag{1.9}$$

Finally, we obtain :

$$\mathrm{Cov}\Big(\mathcal{C}_\ell^{\nu_i\times\nu_j},\mathcal{C}_{\ell'}^{\nu_k\times\nu_l}\Big) = \frac{\delta_{\ell\ell'}}{2\ell+1}\left(\mathcal{C}_\ell^{\nu_i\times\nu_k}\,\mathcal{C}_\ell^{\nu_j\times\nu_l}+\mathcal{C}_\ell^{\nu_i\times\nu_l}\,\mathcal{C}_\ell^{\nu_j\times\nu_k}\right). \tag{1.10}$$

This computation gives the covariance between two cross-spectra, but we also need to account for the limited sky coverage and the binning of spectra. Accounting for these elements, and expressing the covariance of the D_ℓ gives :

$$\mathrm{Cov}\Big(\mathcal{C}_\ell^{\nu_i\times\nu_j},\mathcal{C}_{\ell'}^{\nu_k\times\nu_l}\Big) = \frac{\delta_{\ell\ell'}}{(2\ell+1)f_{\mathrm{sky}}\Delta_\ell}\left(\mathcal{C}_\ell^{\nu_i\times\nu_k}\,\mathcal{C}_\ell^{\nu_j\times\nu_l}+\mathcal{C}_\ell^{\nu_i\times\nu_l}\,\mathcal{C}_\ell^{\nu_j\times\nu_k}\right). \tag{1.11}$$

The resulting covariance matrix is :

$$\boxed{\begin{aligned}\mathrm{Cov}\left(N_\ell^{\nu_i\times\nu_j,\mathrm{sim}},N_{\ell'}^{\nu_k\times\nu_l,\mathrm{sim}}\right) &= \mathrm{Cov}\left(N_\ell^{\nu_i\times\nu_j,\,\mathrm{noise}},N_{\ell'}^{\nu_k\times\nu_l,\,\mathrm{noise}}\right)\\ &\quad+\frac{\delta_{\ell\ell'}}{(2\ell+1)\,f_{\mathrm{sky}}\Delta_\ell}\left(\mathcal{D}_\ell^{\nu_i\times\nu_k}\mathcal{D}_{\ell'}^{\nu_j\times\nu_l}+\mathcal{D}_\ell^{\nu_i\times\nu_l}\mathcal{D}_{\ell'}^{\nu_j\times\nu_k}\right)\end{aligned}} \tag{1.12}$$

1.2 Cosmological parameters' estimation

From the set of auto- and cross-spectra, the associated noise estimation from equation 1.12, and the theoretical model defined in equation 1.1, we perform an MCMC estimation, to estimate the CMB and foregrounds parameters. Since each bandpower averages a large number of nearly independent multipoles within a bin, its distribution is well approximated by a Gaussian [9], justifying the use of a Gaussian likelihood in the parameter estimation that follows.

Remark. A chapter dedicated to sensitivity forecasts for the QUBIC instrument was initially planned as part of this thesis. Its objective was to compare the different reconstruction methods and instrumental configurations under various sky models and simulation settings, in particular the parameters N_{rec} for FMM and N_{freq} for CMM. Unfortunately, due to time constraints, the results are still too preliminary to be included with sufficient confidence. Although the forecasting pipelines are fully implemented, further work is required to optimise their convergence and to achieve a deeper understanding of the obtained results before drawing robust scientific conclusions.

1.2.1 Simulation parameters

In this section, we will use the simulation parameters estimated in Chapter 9. We simulate a full 3 years observation campaign for the QUBIC instrument, using the UWB configuration (see 2.4.2). QUBIC observes its target patch of 15°, located in $(RA = 0°, DEC = 57°)$. We consider a sky composed of only CMB and dust[4], and we fix the amplitude of lensing A_{lens} to 0. In addition, the 147, 213 and 353 GHz bands from Planck are added during the cross-spectra analysis.

1.2.2 FMM cross-spectrum

With the Frequency Map-Making, we are computing frequency maps from QUBIC data. Thanks to spectral imaging, we are reconstructing multiple maps within the physical bandwidth. Increasing the number of maps has the effect of improving the component separation between CMB and astrophysical components, as it increases the number of spectra to fit their associated parameters. Figure 1.3 shows an example of a triangle plot obtained by fitting r, A_d and α_d on the maps cross-spectra. The important values for QUBIC scientific goal are the tensor-to-scalar ratio and its associated uncertainty σ_r. As we are working on simulations, this value has to be compatible with the one used to generate the CMB signal ($r = 0$), while σ_r gives an estimation of the sensitivity of QUBIC instrument.

1.2.3 CMM cross-spectrum

With the Component Map-Making, we are computing component maps from QUBIC data. The idea behind CMM is to perform map-making and component separation at the same time. For this purpose, two algorithms have been developed: Parametric and Blind. In both cases, the output of CMM is one map per component. In our simulation, it is one for CMB and one for Dust. Because of the noise covariance between CMB and dust, due to imperfect component separation, CMB and dust parameters need to be estimated jointly, while being marginalised on the cross-spectrum between CMB and dust, expected to be null. A result of the associated fit is shown in Figure 1.4.

1.2.4 Noise realisations

A key aspect of the noise debiasing procedure is determining the number of noise realisations required to accurately estimate the noise contribution to the angular power spectra. Since there is no straightforward way to predict this number *a priori*, we performed a series of end-to-end simulations using different numbers of noise realisations. For each configuration, the parameter estimation procedure was repeated, and the resulting constraints were analysed. If the number of noise realisations is insufficient, the estimated noise bias may not converge to its true value, potentially leading to biased parameter estimates, including the reconstructed value of r. The objective is therefore to determine the minimum number of realisations for which the recovered parameters become stable.

[4]It is a reasonable assumption to neglect the contribution from synchrotron inside the considered frequency range

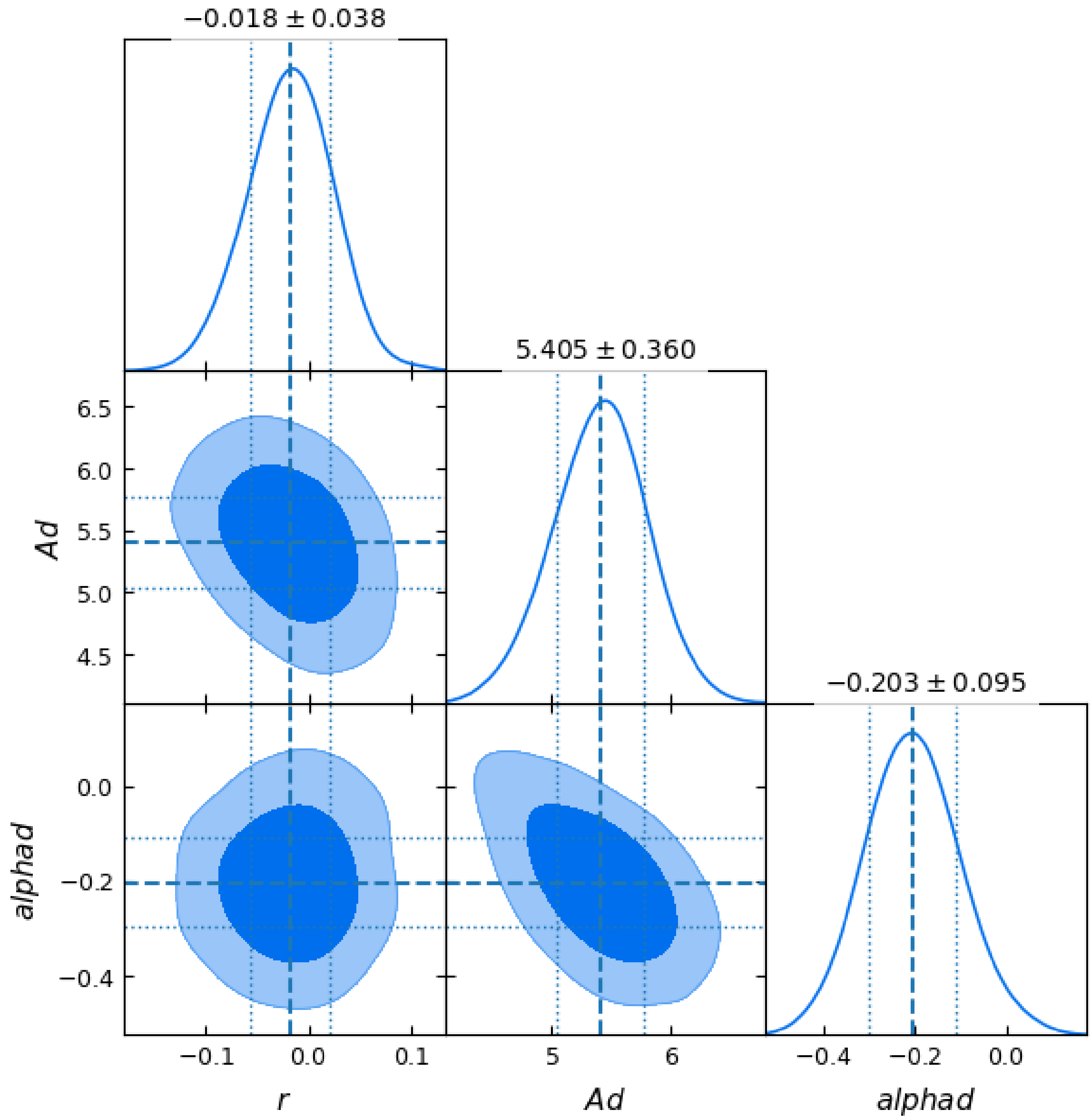

Figure 1.3: Triangle plot of the posterior distribution of the parameters fitted on the cross-spectra obtained with FMM.

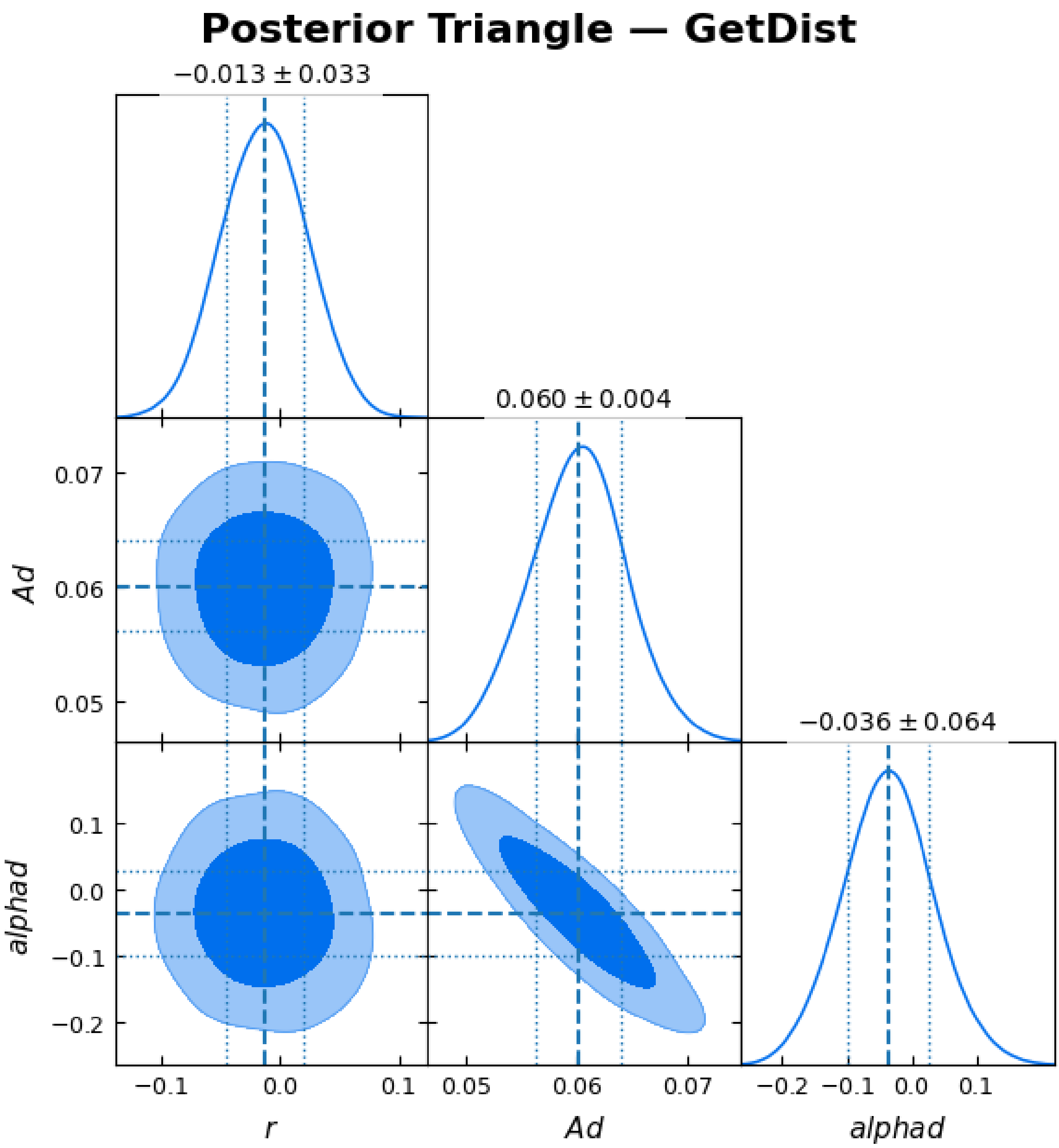

Figure 1.4: Triangle plot of the posterior distribution of the parameters fitted on the cross-spectra obtained with CMM. In this plot, the reference frequency for dust emission is 150 GHz, instead of the most common 353 GHz, hence the small value for A_d.

We observed that the number of noise realisations required for convergence appears to depend on the number of frequency maps included in the fit. A possible explanation is that increasing the number of frequency channels provides additional constraints on the model parameters through a larger set of auto- and cross-spectra. As a result, the fit becomes more sensitive to small inaccuracies in the estimated noise bias. For example, using five frequency maps leads to fifteen auto- and cross-spectra[5], while only a few parameters are fitted. In such a highly constrained configuration, residual biases in the estimated spectra are more easily detected by the likelihood than in a fit involving only two frequency maps. Consequently, a larger number of noise realisations may be required to achieve convergence.

To investigate this effect, we performed the analysis on simulations containing only CMB signal and noise, thereby avoiding additional complications related to component separation. The results are summarised in Figure 1.5. The upper panel shows the fitted value of the tensor-to-scalar ratio r together with its associated uncertainty, obtained from an MCMC analysis. The fit was repeated using different numbers of noise realisations, $N_{\mathrm{real}} = [50, 100, 200, 500, 1000]$. Triangular markers indicate cases where the recovered value is not statistically compatible with the true input value, $r = 0$, revealing a residual bias in the debiasing procedure. The analysis was performed in three different configurations. The blue points correspond to a fit using only the two QUBIC frequency bands, 150 and 220 GHz, in the DB configuration. The orange points correspond to the same QUBIC bands supplemented by the Planck 143, 217, and 353 GHz channels. Finally, the green points include the full set of seven Planck frequency channels in addition to the two QUBIC bands. The lower panel displays the corresponding uncertainty on the fitted tensor-to-scalar ratio, σ_r, as a function of N_{real}. Crosses identify configurations for which the recovered value of r is not compatible with the true input value.

A clear dependence on the number of frequency channels can be observed. The three configurations do not require the same number of noise realisations to achieve a stable and unbiased estimate of r. When only the two QUBIC bands are included, approximately 100 noise realisations are sufficient. Adding the three Planck channels increases this requirement to roughly 300 realisations. Finally, when all nine frequency channels are considered, more than 500 realisations are needed before the recovered value becomes compatible with the input one. These results support the hypothesis that increasing the number of frequency channels makes the fit more sensitive to small inaccuracies in the estimated noise bias.

The observed behaviour could be explained by the fact that the inverse of an unbiased estimator of the covariance matrix is not, in general, an unbiased estimator of the inverse covariance matrix [10]. Consequently, as the dimensionality of the covariance matrix increases, a larger number of realisations is required to obtain an approximately unbiased estimate of its inverse. A more quantitative study of how the required number of realisations scales with the number of frequency bands would be needed to account for this effect efficiently in future forecasts.

[5]The number of spectra scales as $N_{\mathrm{freq}}(N_{\mathrm{freq}} + 1)/2$.

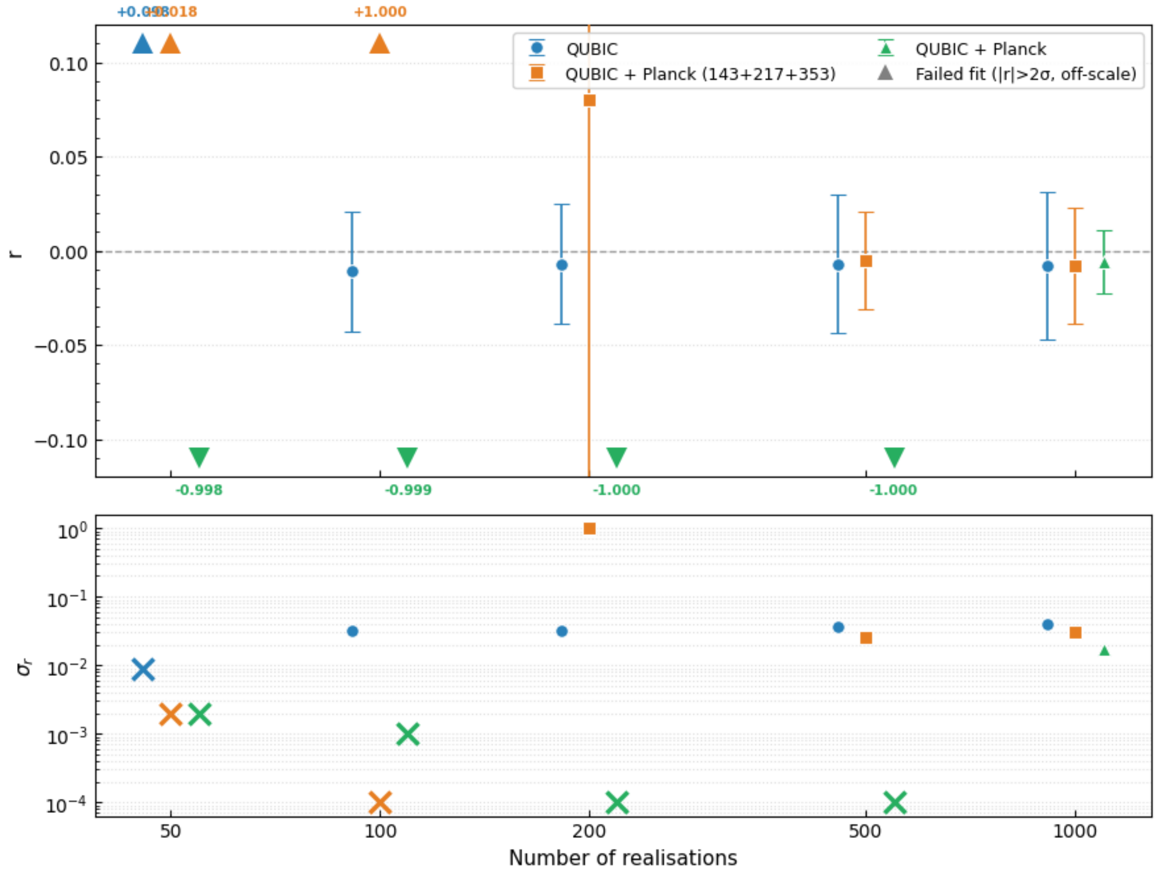

Figure 1.5: Recovered tensor-to-scalar ratio r (top) and corresponding uncertainty σ_r (bottom) as a function of the number of noise realisations used in the debiasing procedure. The three colours correspond to different sets of frequency channels included in the fit. Triangles and crosses indicate configurations for which the recovered value of r is not statistically compatible with the input value $r = 0$.